DATA SCIENTIST CAREER ROADMAP

Student • Python / Java / EDA / Git → Data Scientist • 12-Month Blueprint

■ CURRENT POSITION

Profile: Student | Early-career technologist

Skills: Python, Java, EDA, Git/GitHub

Strengths: Programming foundation, analytical thinking, version control workflow

Edge: Head start over non-technical peers

■ TARGET GOAL

Role: Entry-Level Data Scientist

Deadline: 12 Months

Success Metrics: Portfolio live + Internship/Job offer secured

Salary Range (India): 6–12 LPA entry level

■ SKILL GAP ANALYSIS

Statistics & Probability	Beginner	Intermediate	High	■ Critical
ML Algorithms (sklearn)	Beginner	Proficient	High	■ Critical
Pandas / NumPy	Intermediate	Expert	Medium	■ High
SQL & Databases	Basic	Intermediate	Medium	■ High
Data Visualisation	Basic	Strong	Medium	■ High
Model Evaluation & Tuning	None	Intermediate	High	■ Critical
Storytelling / Communication	Unknown	Strong	Medium	■ High
Deep Learning (basics)	None	Awareness	Low	■ Nice-to-have

■ LEARNING PLAN

Q1 (M1–3): Statistics, NumPy, Pandas mastery, SQL basics

Q2 (M4–6): ML (sklearn), EDA deep-dive, Matplotlib/Seaborn

Q3 (M7–9): Feature Engineering, Model Evaluation, Pipelines

Q4 (M10–12): DL intro, Capstone project, Interview prep

Resources: Andrew Ng ML, Kaggle micro-courses, StatQuest, fast.ai

■ PORTFOLIO PROJECTS

1. **EDA Report** — Kaggle/UCI dataset, insights + visuals
2. **ML Classifier** — Titanic or Heart Disease prediction
3. **NLP App** — Sentiment analyser on product reviews
4. **End-to-End Pipeline** — Data → Model → Streamlit deploy
5. **Capstone** — Domain problem; host on HuggingFace Spaces

■ NETWORKING STRATEGY

• Optimise LinkedIn — headline, banner, featured section

• Post weekly #DataScience learnings (build in public)

• Contribute to open-source GitHub repos (social proof)

• Join Kaggle competitions — aim for Top 30% medals

• Cold-connect 5 DS professionals/week with personal note

■ MONTHLY MILESTONES

M1–2: Stats + Pandas proficiency | Kaggle account active

M3–4: 2 projects on GitHub | 1st LinkedIn post published

M5–6: ML models built | Mid-year skills self-assessment

M7–9: Kaggle Top 30% | 300+ LinkedIn connections

M10–12: 5-project portfolio | Applications live | Offer

■ IMMEDIATE NEXT ACTIONS — Do These This Week

Day 1–2	Day 3–4	Day 5–6	Day 7
Enrol in Andrew Ng ML Specialisation (free audit on Coursera)	Set up Kaggle profile & complete the free Pandas micro-course	Push your first EDA notebook to GitHub on any public dataset	Optimise LinkedIn profile & publish your very first learning post